

VRITHIKA GARG MOHAN

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PROFESSIONAL SUMMARY

Cybersecurity M.S. graduate (GPA 3.9, May 2026) with hands-on experience in security operations, NIST SP 800-53 compliance, and vulnerability assessment. Authored structured System Security Plans and incident documentation aligned with FISMA and FIPS 199/200 standards. Built Python-based automation tools for IOC triage and log analysis, and gained practical SIEM experience with Splunk and Nessus. Seeking a Cybersecurity Analyst or Information Security Analyst role within a compliance-driven or federal environment.

TECHNICAL SKILLS

Security Frameworks: NIST SP 800-53/53A, NIST SP 800-37, NIST SP 800-171, NIST CSF, FISMA, FIPS 199, FIPS 200, ISO/IEC 27001, OWASP Top 10, MITRE ATT&CK

Security Testing & Tools: Nessus, Kali Linux, Wireshark, tcpdump, Nmap, Burp Suite, OWASP ZAP, Metasploit, Snort, Splunk (SIEM), ELK Stack, TryHackMe

Assessment & Documentation: Security Ctrl Assessment, System Security Plans, Requirements Traceability Matrix, Plan of Action and Milestones, Vulnerability Assessment & Management, Risk Assessment, Incident Documentation

Programming & Scripting: Python (NumPy, Pandas, Matplotlib, scikit-learn, regex), SQL, Bash, PowerShell, Java, C, C++, Linux/Unix CLI, Git/GitHub, Data Pipelines

Web & Database: PHP, HTML5, MySQL, Django, REST APIs, JSON

CERTIFICATIONS

- **CompTIA CySA+**
- **Cisco: Introduction to Cybersecurity, Cyber Threat Management, Network Support & Security**

EXPERIENCE

Security Analyst Intern | Tech Fortune, Bengaluru

May 2023 - May 2024

- Maintained structured, reproducible incident logs and audit documentation aligned with NIST SP 800-53A standards, directly supporting compliance tracking and incident response readiness.
- Applied ML-based anomaly detection models to improve alert fidelity, reducing false positives in security monitoring workflows - directly supporting SOC triage accuracy.
- Engineered automated Python data pipelines using Pandas and NumPy to detect and remediate data integrity issues, reducing systemic errors and improving accuracy of downstream security reporting.

PROJECTS

Threat Intelligence Feed Aggregator - IOC Analysis Tool

- Built a Python-based threat intelligence tool that queries AbuseIPDB and AlienVault OTX APIs to assess IP reputation, automating Indicator of Compromise (IOC) lookups across multiple threat feeds simultaneously.
- Implemented a risk scoring engine that classifies IPs as malicious, suspicious, or clean based on abuse confidence scores and historical report frequency, simulating real-world SOC triage and threat hunting workflows.
- Generated structured CSV threat reports with IP reputation scores, country of origin, and abuse categories, cutting manual IOC lookup time from 10-15 minutes to seconds per IP.

Network Traffic Analysis - Academic Security Lab

- Conducted packet capture and protocol analysis using Wireshark to identify anomalous traffic patterns as part of coursework in Managing Cybersecurity Operations.
- Mapped identified attack vectors to defensive strategies, applying real-world threat modeling concepts to reduce mean time to detect (MTTD) anomalous traffic.

Web Vulnerability Scanner

- Built a Python-based web vulnerability scanner that automatically detects security misconfigurations including missing HTTP headers, SQL injection, XSS, and exposed sensitive files across a target website.
- Automated OWASP Top 10 security checks across target sites, streamlining vulnerability discovery typically performed through manual penetration testing.

Log File Analyzer - Security Monitoring Tool

- Developed a Python-based log analysis tool to parse Linux authentication logs, automatically detecting failed SSH login attempts and flagging brute force patterns across system activity.
- Engineered an IP reputation tracker using regex and collections modules to identify and rank suspicious sources by attack frequency, simulating core SOC triage and threat detection workflows.

EDUCATION

M.S. in Cybersecurity - GPA: 3.9 | University of Maryland, Baltimore County

May 2026

Relevant Coursework: Managing Cybersecurity Operations · Global Cyber Capabilities & Trends · Introduction to Cybersecurity

B.E. in Information Science | Vivekananda Institute of Technology, Bengaluru

May 2024

Relevant Coursework: Network Management · Computer Network & Security · Operating Systems · AI & Machine Learning · Data Structures